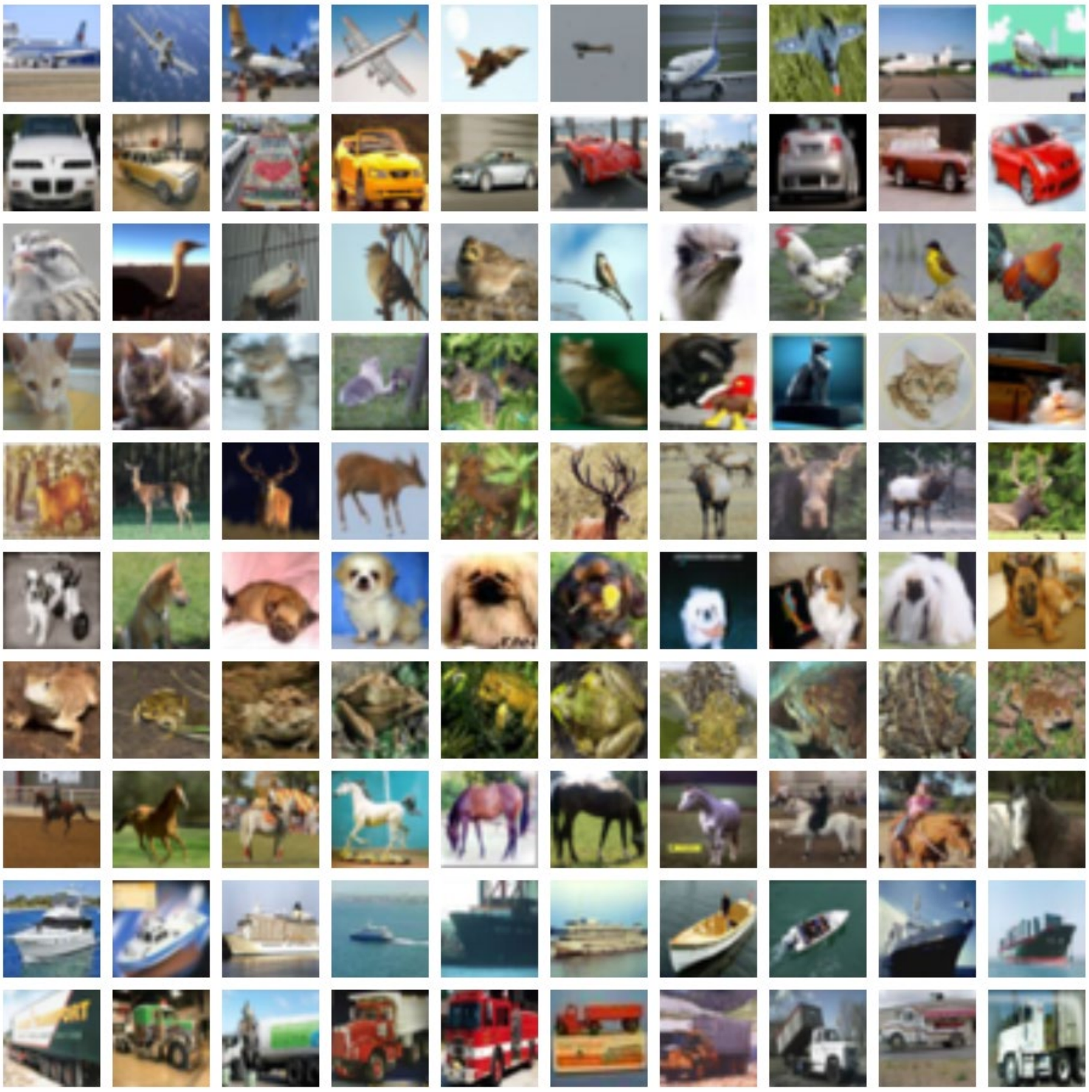

Images from https://www.cs.toronto.edu/~kriz/cifar.html

School Name	School of Computing
Semester	AY2026/27 Semester 1
Course Name	DAAA
Module Code	ST1504
Module Name	Deep Learning

Assignment 2 (CA2: 40%)

The objective of the assignment is to help you gain a better understanding of applying Generative Deep Learning and Reinforcement Learning (RL).

There are three parts in this assignment, Parts A, B, C.

Guidelines

1. You are to work in groups of two for Parts A and B. For Part C, this is an individual task.

2. In this assignment, you will:

   A. Create GAN/VAE for image generation and evaluate the performance of the network.
   B. Create RL to solve the task at hand.
   C. Carry out some literature research and prepare a technical paper on CNN, RNN, GAN/VAE or RL. (Just ONE topic.)

3. For Parts A and B, you should prepare the following:

   a) Jupyter notebook including your code, comments and visualisations (.ipynb).
   b) In addition, please save a copy of the Jupyter notebook as a .html file.
   c) Include your best neural network weights (.h5 file).
   d) A deck of presentation slides (.pptx file) for your project.
   e) A statement indicating the contributions by each member of the group, including percentage of workload for specific contributions.

   *NOTE: Each student must be well-versed with **both** Parts A and B. Whilst you can consider splitting the workload, you will still be evaluated on everything in this CA. You cannot expect to only do one part but know little or nothing about the other part.*

4. For Part C, you should submit your file as a Word (.docx) or .pdf document. If you write your own codes, you must include them.

5. Submit all materials in a zipped file. Each student must submit his/her own complete set of files, even if you are sharing for Parts A and B.

   Any missing files (like missing .h5 file for best weights, missing .html file for your code, etc.) would incur marks deduction.

6. The normal SP's academic policies on Copyright and Plagiarism applies. Please note that you are to cite all sources. You may refer to the citation guide available at: https://sp-sg.libguides.com/citation

7. You need to submit your declaration of academic integrity. You may access this document on Brightspace. Without this, your submission is deemed incomplete.

Submission Details

Deadline: 11 Aug 2026, 08:00 AM
Submit through: Brightspace

Late Submission

50% of the marks will be deducted for assignments that are received within ONE (1) calendar day after the submission deadline. No marks will be given thereafter. Exceptions to this policy will be given to students with valid LOA on medical or compassionate grounds. Students in such cases will need to inform the lecturer as soon as reasonably possible. Students are not to assume on their own that their deadline has been extended.

Neural network models

You must build your own neural network models, with explanations and justifications.

Your neural network models can be improved upon with tweaks to your architectures.

If you wish to implement transfer learning, it is only acceptable after you have done the above (building your own models with justifications).

Otherwise, transfer learning is rejected.

Save the best weights of your neural networks. This is important for reproducibility without having to re-train over some extended duration.

Reminder: Please check that all files are valid, especially after zipping. If files cannot be opened, it would be considered as no submission. It is your responsibility as students to ensure this is properly carried out.

Learning outcomes

1) Create generative deep learning for creative artificial intelligence.
2) Create deep reinforcement learning for agents solving some desired tasks.
3) Explain advanced deep neural network architectures within the context of image classification, natural language processing, generative deep learning, and reinforcement learning.

PART A: GAN/VAE (45 marks)

Task

Apply some suitable GAN/VAE architectures to the problem of image generation. You can use either GAN or VAE, or both. Use the given dataset for CIFAR10 to create **1000** small images. There should be 10 classes of images for you to generate. You **must submit** your generated images (otherwise, there would be marks deduction).

You should implement some ways to evaluate the quality of your images:

A simple "by-eye" or "eye-power" method is to just generate say 100 images, and count manually how many are acceptable images (not perfect, but display some plausible features of the images --- perhaps as "clear", "marginal", "nonsense"). Alternatively, you can consider other metrics/indicators. (Optional metrics: Fréchet Inception Distance, Inception Score.)

If you are asked to generate images of a specific class, propose a way of doing it.

If you are asked to generate black-and-white images instead of coloured ones, do you think it would be easier or harder to produce better quality results?

What class(es) is/are relatively easier/harder to generate? Why?

Note: Do not worry too much about generating "perfect-looking images". Whilst quality of images would represent the quality of your model, the more important aspects are the process, your workflow and planning, your EDA, evaluation analysis, etc., as a whole.

As this is a pair-work, you should discuss and optimise the GAN/VAE training between yourselves, making use of available computing resources for both of you.

Dataset

Use the CIFAR10 dataset:

tf.keras.datasets.cifar10.load_data()

You **cannot use any external data** to train your model.

Nevertheless, you are allowed to apply augmentation on the provided data, if you wish. If you choose to do so, you must concretely explain why you make such a choice, as well as investigate whether it is actually beneficial.

Submission requirements for Part A

1. Submit a zip file containing all the project files (source code, Jupyter notebook .ipynb file, .html file, and best neural network weights .h5, slides, 1000 generated images).
2. Submit a .docx file containing the list of specific contributions by each team member in the deliverables for Part A.
3. Submit online via the Assignment link.

Evaluation criteria:

Component	Description	Marks
Background research, feature engineering	Provide background of the problem. Explain the approach(es) being taken to solve the problem. Some feature engineering on the provided dataset may be carried out (if desirable).	9
Application of GAN/VAE, explanation of architecture(s)	Describe the GAN/VAE framework(s) being used in this problem. Explain clearly why such GAN/VAE framework(s) can be used and how it can solve this problem.	9
Evaluation and improvements of GAN/VAE performance	Evaluate the performance of the GAN/VAE model in this problem. Improve the GAN/VAE model performance systematically.	9
Presentation/Demo	Give a good, clear presentation strictly within the allocated time. Quality of presentation slides should be engaging and effectively helps with communicating the key points.	9
Quality of report (Jupyter)	Jupyter notebook should be well-documented with comments and markdowns to explain the codes. It should also be well-organised.	9

Note: You are building deep neural networks, which are remarkable things that in this case, produce creative outputs. Think about how long it took top tech companies to train state-of-the-art models, like ChatGPT, Co-Pilot, DeepSeek, etc., with massive computing power. In this exercise, you have the opportunity to build your very own such generative AI models.

Yes, it will take time to train, but do enjoy the process and your results. Make use of the time in the lab to train your models. Do not worry too much about final results, as the more important thing is your workflow on building the models.

PART B: REINFORCEMENT LEARNING (45 marks)

Task

Apply a suitable modification of deep Q-network (DQN) architecture to the problem. Your model should exert some appropriate torque on the pendulum to balance it.

<u>You must use DQN and satisfactorily demonstrate its viability.</u>

(Note: The action space here is continuous. Hence, you should consider discretising it in order to apply DQN.)

Here are several setups that you must investigate (set different values of g when creating the "env" for the pendulum):

1) The default strength for gravity, no need to change g.
2) Free-fall, $g = 0$.
3) Anti-gravity, $g = -10$.
4) Supergravity, $g = 15$.

In your work, you should plan clearly what you are doing, your approaches, and how you systematically optimise your solutions.

For example, what hyperparameters can you tune? Is one trial enough or should you repeat the trials? Why?

How do you conclusively demonstrate your so-called "best setup" to be the best? Are you considering fastest learning, most stable learning, or some other criteria that you choose to define?

<u>Optional</u>: You may consider other reinforcement learning architectures, if you wish, but only after successfully implementing DQN. Otherwise, any other non-DQN architecture will be rejected.

Dataset

Please use the following environment from OpenAI Gym.
https://www.gymlibrary.dev/environments/classic_control/pendulum/

NOTE: Stay within the older version of gym 0.17.3, as implemented in the lab for cartpole.

Submission requirements for Part B

1. Submit a zip file containing all the project files (source code, Jupyter notebook .ipynb file, .html file, and best neural network weights .h5, slides).
2. Submit a .docx file containing the list of specific contributions by each team member in the deliverables for Part B.
3. Submit online via the Assignment link.

Evaluation criteria:

Component	Description	Marks
Background research, explanations of approach(es) taken	Provide background of the problem. Explain the approach(es) being taken to solve the problem.	9
Application of RL with appropriate rationale and explanations	Describe the RL framework being used in this problem. Explain clearly why this RL framework can be used and how it can solve this problem.	9
Evaluation and improvements of RL	Evaluate the performance of the RL agent in this problem. Improve the RL agent's performance systematically.	9
Presentation/Demo	Give a good, clear presentation strictly within the allocated time. Quality of presentation slides should be engaging and effectively helps with communicating the key points.	9
Quality of report (Jupyter)	Jupyter notebook should be well-documented with comments and markdowns to explain the codes. It should also be well-organised.	9

PART C: Technical Paper (10 marks)

This part of the assignment is to be completed individually. This is a challenge task for students who wish to attempt it for higher marks.

Write a technical paper in single column format on any **ONE** of the following topics:

- CNN (computer vision, in general)
- RNN (natural language processing, in general)
- GAN, VAE (generative AI, in general)
- RL (reinforcement learning, in general)

In our lessons, we covered some fundamental and basic architectures for these topics. However, there are certainly more advanced models available beyond what we covered. Take this opportunity to dive deeper into such extensions.

Here are some examples (but you are free to do other topics not listed here):

There is a raft of advanced CNN models: ResNet, Inception, VGG. Consider focusing on one such interesting models to you, and explore deeper.

For RNN, you may delve into how ChatGPT works, the underlying architecture, large language models, for example.

For GAN, there are various other improvements apart from the basic DCGAN. Even VAE has extensions to areas like physics.

For RL, we only covered DQN. How can we improve DQN? Double DQN, duelling, priority sampling, etc. How about other RL architectures beyond DQN?

The paper should have the following components:

1. Abstract
2. Introduction
3. Related Works
4. Dataset/Methodology/Experiment
5. Discussion
6. Conclusions
7. References

Submit the paper in Word or PDF format (page limit of 10 pages). If you write your own codes, you must submit them to support your work.

NOTE: Do not just copy some advanced research paper without some proper understanding. It does not need to be too technical/mathematical.

It is more meaningful to relate the advance topics you are exploring to something we dealt with in class and in CA1/CA2. Demonstrate how it improves upon those basic architectures that we covered.

Do not provide a superficial review or “name-dropping” of advanced terms without deeper understanding.

Evaluation criteria (total 10 marks):

Component	Description
Review of research articles	Provide reviews of relevant research articles used in relation to your technical paper. Citations of those research articles must be properly included. Note: The reviews must be relevant and meaningful.
Dataset	Introduce the dataset to be used. Carry out relevant preprocessing in preparation for model building.
Models	Build an original/advance model, with some interesting results. Include submission of Jupyter notebook.
Elaboration, evaluation of models, discussions	Demonstrate efficacy of model results in relation to the topic being discussed. Relate with the research articles that are reviewed earlier.
Appropriate level of topic	The topic should be slightly beyond what is covered in class. However, it should be relatable to the fundamental topics covered in class. Explain how this topic relates and advances the fundamental topics covered in class. The topics should not be overly technical with “name-dropping” of advanced terms that does not demonstrate deeper understanding.

Note: These are the five key evaluation criteria for Part C. The overall coherence of the technical paper will add up to a total of 10 marks, based on these five criteria.

— *End of Assignment* —